

Question No:4

Question description

Byteland has n cities and m roads between them. The goal is to construct new roads so that there is a route between any two cities. Your task is to find out the minimum number of roads required, and also determine which roads should be built.

Constraints

$$1 \leq n \leq 10^5$$

$$1 \leq m \leq 2 \cdot 10^5$$

$$1 \leq a, b \leq n$$

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

After that, there are m lines describing the roads. Each line has two integers a and b : there is a road between those cities.

A road always connects two different cities, and there is at most one road between any two cities.

Output

First print an integer k : the number of required roads.

Then, print k lines that describe the new roads. You can print any valid solution.

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4 2
1 2
3 4
```

EXPECTED OUTPUT

```
1
2 4
```

Test Case 2

INPUT (STDIN)

```
4 2
2 3
2 4
```

EXPECTED OUTPUT

```
1
1 4
```

Code:

```
1  #include <stdio.h>
2
3  #define MAXN 100005
4
5  int parent[MAXN];
6  int rank[MAXN];
7
8  /* Find the parent of a city */
9  int find(int x)
10 {
11     if (parent[x] != x)
12         parent[x] = find(parent[x]);
13
14     return parent[x];
15 }
16
17 /* Join two cities */
18 void unite(int a, int b)
19 {
20     a = find(a);
21     b = find(b);
22
23     if (a == b)
24         return;
25
26     if (rank[a] < rank[b])
27     {
28         parent[a] = b;
29     }
```

```
30     else if (rank[a] > rank[b])
31     {
32         parent[b] = a;
33     }
34     else
35     {
36         parent[b] = a;
37         rank[a]++;
38     }
39 }
40
41 int main()
42 {
43     int n, m;
44
45     scanf("%d %d", &n, &m);
46
47     /* Initially every city is its own component */
48     for (int i = 1; i <= n; i++)
49     {
50         parent[i] = i;
51         rank[i] = 0;
52     }
53
54     /* Read existing roads */
55     for (int i = 0; i < m; i++)
```

```

56     {
57         int a, b;
58         scanf("%d %d", &a, &b);
59
60         unite(a, b);
61     }
62
63     /*
64      Store one representative city
65      from every connected component.
66     */
67     int representatives[MAXN];
68     int count = 0;
69
70     for (int i = 1; i <= n; i++)
71     {
72         if (find(i) == i)
73         {
74             representatives[count++] = i;
75         }
76     }
77
78     /*
79      We need count - 1 new roads.
80      Connect every component to the first component.
81     */
82     printf("%d\n", count - 1);
83
84     for (int i = 1; i < count; i++)
85     {
86         printf("%d %d\n", representatives[0], representatives[i]);
87     }
88
89     return 0;
90 }

```

Output:

Test Case-1:

```
DS\dsa\dsa\dsa
4 2
1 2
3 4
1
1 3
PS C:\Users\kesha\OneDrive\Documents>
```

Test Case-2:

```
DS\dsa\dsa
4 2
2 3
2 4
1
1 2
PS C:\Users\kesha\OneDrive\Documents\3sem\DSA\DSA Practice>
```

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